

## **Reading skill and the use of orthographic knowledge by Mature Readers**

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**Summary.** Butler and Hains (1979) found individual differences in the effect of word length on RT in word naming and lexical-decision tasks; subjects scoring high on a vocabulary test were much less affected by word length than subjects with low scores. The present study attempted to determine whether such differences could be due to differences in the use of intra-word structure, since word recognition has been linked, both empirically (Mason, 1978; Scheerer-Neumann, 1981) and theoretically (Mewhort & Campbell, 1981) to the use of orthographic redundancy. In Experiment 1, subjects were shown tachistoscopically presented letter strings varying in approximation to English. High vocabulary subjects were more capable of capitalizing on the redundancy with the higher-order approximations. Experiment 2 then established that performance on the vocabulary test was correlated with reading skill, measured by the Nelson-Denny Reading Test, and that skilled readers were more sensitive to the syllable structure in words. In the experiment, subjects were shown eight-letter words presented as three separate word units which either matched or violated syllable boundaries in the word. Skilled readers were more sensitive to the difference between syllable and nonsyllable units than were less skilled readers. The results obtained by Butler and Hains appear to reflect differences in reading skill and the efficiency of letter sequence parsing prior to word recognition.

The systematic investigation of individual differences can contribute a great deal to our understanding of word recognition and reading processes. A comparison of good and poor readers can identify subskills necessary for rapid and efficient processing. Because of the educational implications, such studies often focus on younger chil-

dren at the grade-school level, usually including an undifferentiated group of adults as a control. However, these studies need not be restricted to children. There is ample evidence of substantial individual differences among adults (e.g., Mason, 1978, 1980; Massaro & Taylor, 1980).

Butler and Hains (1979) found individual differences in word recognition performance that were linked to a more general measure of verbal ability. Subjects were tested on both word-naming and lexical-decision tasks and, in addition, were given the informal but quite demanding test of English vocabulary. Reaction time (RT) for both tasks increased as a function of word length but, more importantly, the college students who achieved high scores on the vocabulary test were less affected by word length than were students with low scores. These results suggest that at some stage processing time depends on word length and at this stage the High Vocabulary subjects are either more efficient or faster than the Low Vocabulary subjects. The simplest interpretation, and at the same time the most vague, is that the High Vocabulary subjects "take a more wholistic approach to word recognition" (Taylor & Taylor, 1983, p. 200).

Similar effects have been obtained in studies with young children and in other studies with adults. Perfetti, Finger, and Hogaboam (1978) found that highly skilled grade-three readers produced shorter vocalization latencies in a word-naming task and were less affected by the number of syllables in a word than were less skilled readers. The same pattern of results emerged for grade-five readers (Perfetti & Hogaboam, 1975) and persisted even when the words were presented in a discourse context (Perfetti, Goldman, & Hogaboam, 1979). Mason (1978) reported similar differences in vocalization latencies among college students with four- and six-letter units; Mason, Pilkington, and Brandon (1981) replicated the effects with words and showed that the same effects could be obtained with number sequences. Both the results with children and the results with adults have been explained in similar ways. Perfetti, Finger, and Hogaboam (1978, p. 730) claimed that: "Inefficiency in alphabetic verbal coding rather than use of information constraint or word retrieval seems to be the major source of reader differences in vocalization latencies." Mason (1978, p. 568) suggested that skilled readers "may make more use of variables of orthographic regularity."

Jackson (1980) has proposed that the principal difference between skilled and less skilled readers lies in the speed with which they are able to access relevant memory codes. In a series of studies, Jackson (Jackson & McClelland, 1975, 1979; Jackson, 1980) found that skilled readers are reliably faster than less skilled readers at any task requiring memory retrieval and the difference in speed depends on the complexity of the memory code. The individual differences noted by Butler and Hains could reflect speed differences in retrieving orthographic information during word processing or speed differences during lexical retrieval.

The present study does not deal with potential speed differences but does examine the possibility that the results found by Butler and Hains may reflect different strategies for using orthographic knowledge. The effective use of orthographic knowledge is a central feature of a scan-parse model proposed by Mewhort and Campbell (1981). Within this model, the chief limits on performance do not involve initial letter identification nor subsequent lexical retrieval but instead involve the transfer

of letter information to short-term memory. This transfer entails a conversion from a spatial to a temporal code and is accomplished by a left-to right directional scanning process. Considerable independent evidence exists for such a mechanism (see Mewhort & Campbell, 1981, but also Scheerer, 1972, 1973; Butler, 1979; Tramer, Butler, & Mewhort, in press). In order to facilitate transfer, the scanning mechanism utilizes orthographic information to parse letters into higher-order units, based on the syllable (Mewhort & Beal, 1977), so that there are fewer individual units to transfer. The scan-parse mechanism, as described by Mewhort and Campbell, is so dependent on parsing that *negative* orders-of-approximations, that is, letter sequences that consistently violate rules, will yield poorer performance than random sequences (Campbell & Mewhort, 1980).

The individual differences found by Butler and Hains are quite consistent with the scan-parse model. The High Vocabulary subjects may make greater use of orthographic knowledge to parse the letters in a word into larger functional units, than do the Low Vocabulary subjects. For a word of any given length, the High Vocabulary subjects would have a distinct advantage of having fewer units to transfer to short-term memory and this advantage would increase with longer words. This is quite consistent with the notion that the High Vocabulary subjects are using a "more wholistic" strategy but is a much more precise statement. The relationship between knowledge of English vocabulary and parsing may be somewhat indirect in that parsing strategy may affect reading skill which, in turn, determines the quantity and complexity of material read and, hence, affects one's performance on a vocabulary test.

This parsing hypothesis would also account for the results obtained by Perfetti and his colleagues and Mason (1978). It will not account for the speed differences found by Jackson (1980) but these may reflect speed differences in mechanistic processes correlated with general verbal ability (Hunt, 1978).

### Experiment 1

If the parsing hypothesis is correct, the High and Low Vocabulary subjects should show substantial differences on a tachistoscopic recognition task using different orders-of-approximations to English (Campbell & Mewhort, 1980). Scheerer-Neumann (1981) reported such effects with good and poor readers among third-grade German school children but there is some question of whether such differences are found with adults. Massaro and Taylor (1980) did not find any differences between good and poor readers at the college level, while Mason (1978) did find individual differences in terms of the utilization of orthographic structure.

The present experiment tested the efficiency of processing for High and Low Vocabulary subjects using a tachistoscopic recognition task with eight-letter pseudo-words of zero-, second-, and fourth-order approximation-to-English (Hirata & Bryden, 1971). Such stimuli differ in the extent to which the letter sequences reflect the sequential constraints of printed English. Zero-order approximations are essentially random letter sequences while fourth-order approximations, such as DINGLECT or POLICKET, resemble real words and are generally pronounceable.

## Method

*Subjects.* Twenty-four subjects were selected at random from an Introductory Psychology class at Queen's University, all members of which had previously been given the vocabulary test described in the previous experiments. Scores for the subjects ranged from 16 to 34 with a median of 25.5. Subjects were divided into High and Low Vocabulary groups on the basis of a median split. The mean score for the Low group was 21.33 and the mean for the High group was 28.75.

*Materials.* The stimuli consisted of 30 eight-letter sequences of zero-order approximation to English (OAE), 30 of second-OAE, and 30 of fourth-OAE. No letters were repeated within a sequence. The letter sequences were selected from the lists prepared by Hirata and Bryden (1971). Each letter sequence was printed horizontally on white cardboard using black capital Letraset Instant Lettering (No. 287). As viewed by the subjects, each stimulus subtended about  $3.15^\circ$  horizontally and  $0.45^\circ$  vertically. An additional set of 10 zero-order sequences were prepared as practice stimuli. A masking stimulus was prepared using a portion of a photograph of random black and white squares (Gregory, 1966, p. 59), that completely covered the area occupied by each letter array.

The stimuli were presented using a Gerbrands Harvard Tachistoscope (model – T4A) with Sylvania F4T4/CWX bulbs.

*Procedure.* Each trial began with a verbal "Ready" signal from the experimenter, followed by a 100 ms presentation of a stimulus. Subjects were asked to report as many letters as possible in any order they wished, and to guess if they were uncertain. Each stimulus was illuminated at approximately  $15 \text{ cd/m}^2$  and was followed, either immediately or after a 60 ms delay, by the visual noise mask presented for 1000 ms at  $17 \text{ cd/m}^2$ . In the delay-of-mask condition, a plain white field was presented for the 60 ms between stimulus and mask and was illuminated at  $15 \text{ cd/m}^2$ .

Each subject viewed the 90 stimuli twice, once with no delay-of-mask and once with the 60 ms delay. Half of the subjects received the no delay trials first while the remainder received the delayed trials first. Within each block of trials the order of stimuli were randomized so subjects could not anticipate the OAE of the stimulus on each trial. Prior to each block of 90 trials, the subjects were given 10 practice trials under the same viewing conditions as the block of trials about to be presented.

The record of each subject's performance included the letters reported on each trial and the order in which those letters were reported.

## Results and Discussion

A response was considered correct if it matched one of the letters in the display, regardless of the position of the response in the response sequence. Two sets of scores were then derived for each subject. One score was the *number of correct responses*, summed over 30 trials, for each stimulus position in the zero-, second-, and fourth-OAE sets. The second score was the *run length* for each trial (Campbell & Mewhort,

1980). This was defined as the number of consecutive correct responses beginning with the left-most letter and proceeding uninterrupted toward the right-most letter. Any error of omission or commission terminated the run for a response sequence. Run length is a measure of the number of correct responses and the extent to which the subject uses a strict left-to-right order for reporting the stimuli. Although the order of report was not specified, the subjects generally report in a left-to-right order.

Both sets of scores were submitted to analysis of variance in order to determine the effects of stimulus position, OAE, and vocabulary for the number of correct responses and the effects of OAE and vocabulary for run length. The analyses were run twice, once with subjects divided into High and Low Vocabulary groups based on a median split and once with subjects divided into High, Medium, and Low Vocabulary groups of eight each. The analyses lead to the same conclusions whether subjects were split into two or three groups so the discussion will refer only to the first analysis with the median split.

The overall results, with number of correct responses transformed to probability of a correct response, are shown in Table 1. Performance increased with OAE,  $F(2,44) = 116.04$ ,  $p < 0.001$ , and with greater delay of mask,  $F(1,22) = 48.87$ ,  $p < 0.001$ , and was more affected by OAE at the longer mask delay,  $F(2,44) = 4.74$ ,  $p < 0.02$ . Overall, High and Low Vocabulary subjects recalled about the same number of letters correctly,  $F(1,22) = 0.03$ , but the High Vocabulary subjects were more affected by OAE than were the Low Vocabulary subjects,  $F(2,44) = 6.28$ ,  $p < 0.01$ . Delay of mask affected both vocabulary groups,  $F(1,22) = 3.4$ ,  $p > 0.05$  but there was no evidence of a three-way interaction,  $F(2,44) = 0.36$ , with OAE.

The principal difference between the High and Low Vocabulary subjects is illustrated in Figure 1. This figure shows mean accuracy for each stimulus position in the three OAE sequences. The results reflect the pattern of means in Table 1; Low Vocabulary subjects recalled more than High Vocabulary subjects from the zero-order sequences while High Vocabulary subjects recalled more than Low Vocabulary subjects from the fourth-order sequences. In general, performance decreased from left-to-right across the display,  $F(7,154) = 95.88$ ,  $p < 0.001$ , but this decrease was less for the higher OAE sequences,  $F(14,308) = 12.85$ ,  $p < 0.001$ . This result, however,

Table 1. Mean probability of a correct response collapsed across stimulus position

| Vocab.<br>Group | Mask<br>Delay | Order of Approximation |        |        | $\bar{x}$ | $\bar{x}$ |
|-----------------|---------------|------------------------|--------|--------|-----------|-----------|
|                 |               | zero                   | second | fourth |           |           |
| HIGH            | 0             | 0.34                   | 0.50   | 0.59   | 0.48      | 0.52      |
|                 | 60            | 0.41                   | 0.56   | 0.69   | 0.56      |           |
|                 | $\bar{x} =$   | 0.38                   | 0.53   | 0.64   |           |           |
| LOW             | 0             | 0.38                   | 0.48   | 0.52   | 0.46      | 0.53      |
|                 | 60            | 0.50                   | 0.62   | 0.68   | 0.60      |           |
|                 | $\bar{x} =$   | 0.44                   | 0.55   | 0.60   |           |           |

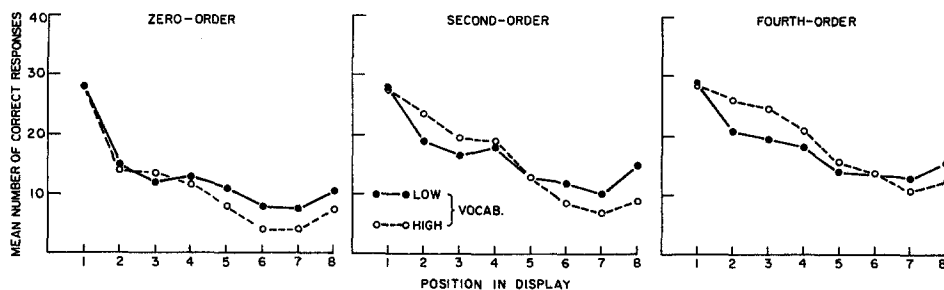


Fig. 1. Mean number of correct responses for each stimulus position within zero-, second-, and fourth-order approximations to English for High and Low Vocabulary groups in Experiment 1.

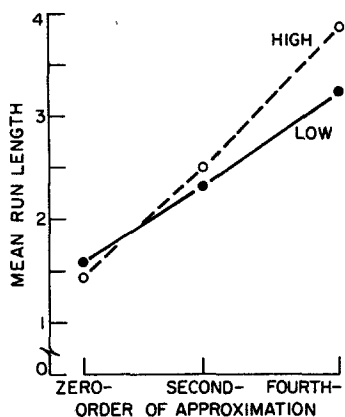


Fig. 2. Mean run length as a function of order of approximation to English for High and Low Vocabulary groups in Experiment 1.

was stronger for the High Vocabulary subjects than for the Low Vocabulary subjects. High Vocabulary subjects showed a much greater left-to-right difference than did Low Vocabulary subjects,  $F(7,154) = 4.12$ ,  $p < 0.001$ , and this difference changed as OAE increased,  $F(14,308) = 3.06$ ,  $p < 0.001$ . High Vocabulary subjects appeared to concentrate on the left-most items with the zero-order sequences and to succeed in moving more to the right with the second-order than fourth-order sequences. The Low Vocabulary subjects, on the other hand, appear to recall more from the right end than do the High Vocabulary subjects and seem to show a benefit of OAE across all stimulus positions.

Mean run length scores for each group are shown in Figure 2 and these confirm the effects noted in Figure 1. Run length, that is, the number of consecutive correct responses beginning with the leftmost letter, was affected by OAE,  $F(2,44) = 80.29$ ,  $p < 0.001$ , and by delay of mask,  $F(1,22) = 50.0$ ,  $p < 0.001$ . Mean run length was 2.62 for the High Vocabulary subjects, compared to 2.37 for the Low Vocabulary subjects,  $F(1,22) = 0.41$ , but the High Vocabulary subjects showed a greater effect of OAE on run length. As Figure 2 shows, run length is a linear function of OAE and the increase is greater for the High Vocabulary subjects,  $F(1,44) = 5.66$ ,  $p < 0.02$ , than for the Low Vocabulary subjects.

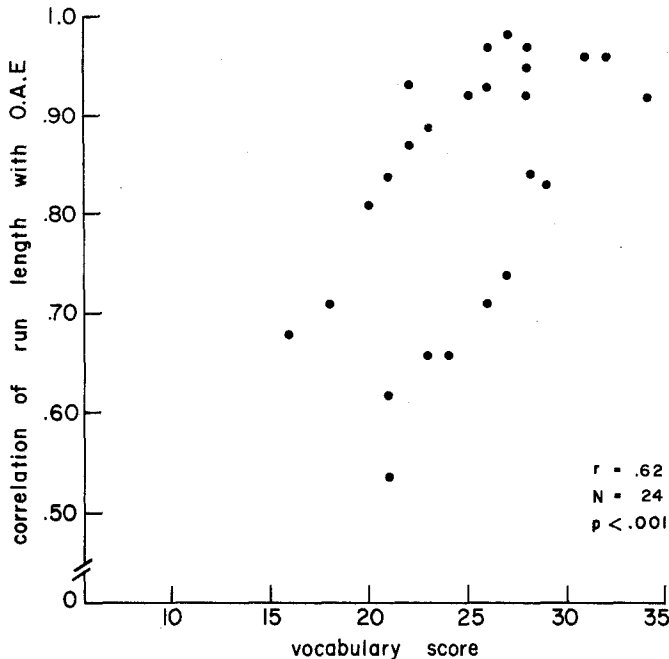


Fig. 3. For each subject in Experiment 1, the Pearson product-moment correlation between run length and order of approximation is shown as a function of the subject's vocabulary score.

The relationship between the subject's vocabulary score and the effects of OAE are further confirmed by Figure 3. For this figure, the Pearson Product-Moment correlation between run length and OAE was calculated for each individual subject and the resulting correlation coefficients are represented as a function of the subjects' vocabulary scores. Subjects with the higher vocabulary scores clearly show a greater effect of OAE on run length,  $r(22) = 0.62$ ,  $p < 0.01$ .

The individual differences in the effects of word length found by Butler and Hains appear to involve differences in the use of orthographic structure during word recognition. The present study quite unequivocally shows that the High Vocabulary subjects are more sensitive to orthographic structure than are the Low Vocabulary subjects. The results are quite consistent with the notion that High Vocabulary subjects make greater use of orthographic knowledge to parse letter sequences into larger units prior to transfer to short-term memory and, hence, are less sensitive to actual word length.

It would also appear that differences in vocabulary size may be quite incidental to the differences in reading skill. This study, unlike the previous study, used pseudo-words which do not correspond to actual lexical entries, so the size of the subject's vocabulary is quite irrelevant. It seems likely, however, that vocabulary is affected by reading experience. Experience, in turn, is probably determined by reading skill and reading skill appears to be the important discriminating factor with these subjects.

From a historical point of view, it is interesting to note that the present results are neither all that new or surprising. Hoffmann (1927) was well aware that the recognition span for familiar letter sequences increased with reading experience and reading skill. Furthermore, the data reported in the leftmost panel of Figure 1 is very similar to results reported by Crosland (1939) with children classed as good or poor readers. In citing Crosland's work, Anderson and Dearborn (1952, p. 224) discuss the importance of "left-to-right mindedness" in skilled word recognition.

## Experiment 2

According to the scan-parse theory (Mewhort & Campbell, 1981) a knowledge of English orthography allows the reader to parse a letter string into higher-order units and these higher-order units are specifically identified as syllables. The importance of the syllable as a unit in identifying words was demonstrated in a series of studies by Mewhort and Beal (1977). In one of the experiments, subjects were asked to identify eight-letter words, shown on a computer-controlled CRT monitor, but the subject was not shown all the letters at the same time. Each word was divided into three units that corresponded to syllables (e.g., AD/DI/TION and HOS/PI/TAL) or violated syllable boundaries (e.g., ADD/IT/ION and HO/SP/ITAL). Subjects were shown only one unit at a time with an inter-unit interval, IUI, ranging from 0 to 625 ms. Each unit occupied a position on the screen appropriate to its actual position in the word but the units could be shown in either a left-to-right or right-to-left order.

The logic of the experiment was relatively straightforward. With no pause between units the letters would appear to be simultaneous and subjects should experience little difficulty in identifying the words. With a slower presentation rate, that is longer IUIs, subjects should again experience relatively little difficulty if the units form syllables. In this case, "short-term memory merely has to queue the syllable units" (Mewhort & Beal, 1977, p. 635). With syllables, parsing is accomplished by the mode of presentation. Problems should arise, however, with the nonsyllable units. When the units violate syllable boundaries, the slower rates of presentation should force subjects to queue the units then reparse them in short-term memory before recognizing the word. The extra effort and unusual reparsing requirement should increase the error rate considerably. The results obtained by Mewhort and Beal confirmed a substantial difference between syllable and nonsyllable units and a considerable effect of direction of presentation.

The results of Experiment 1 suggest that there are individual differences in sensitivity to orthographic redundancy. If this is the case, subjects most sensitive to orthographic redundancy should also be most sensitive to the syllable/nonsyllable difference because these subjects are most likely to use the syllable as the higher-order unit when parsing. Subjects less sensitive to redundancy must use a parsing unit smaller than the syllable and will be less sensitive to the syllable/nonsyllable difference. Mewhort and Beal were concerned only with the strategy used by the skilled reader and assumed that university students were a reasonably homogeneous population. Experiment 1, however, suggests individual differences within their paradigm, and those differences should be related to reading skill.



The present experiment, then, is a direct attempt to examine parsing strategy using the same procedure as Mewhort and Beal. This experiment does depart from the previous experiment and the study by Butler and Hains by using subjects classed as Good and Poor readers, rather than classed according to performance on the vocabulary test. This was done for two reasons: first, the vocabulary test was an informal test with unknown reliability and validity; second, Experiment 1 has shown that the primary discrimination among the subjects is in terms of word recognition skill and vocabulary size is an incidental correlate. To determine whether vocabulary knowledge and reading skill were correlated, we administered both the vocabulary test (described by Butler & Hains, 1979) and the Nelson-Denny Reading Test (Form B; Brown, 1960) to a group of 48 second-year university students. The results of the testings are shown in Figure 4 and Table 2. Two subsamples of subjects were selected for the present experiment and their scores are indicated on Figure 4. These two groups of

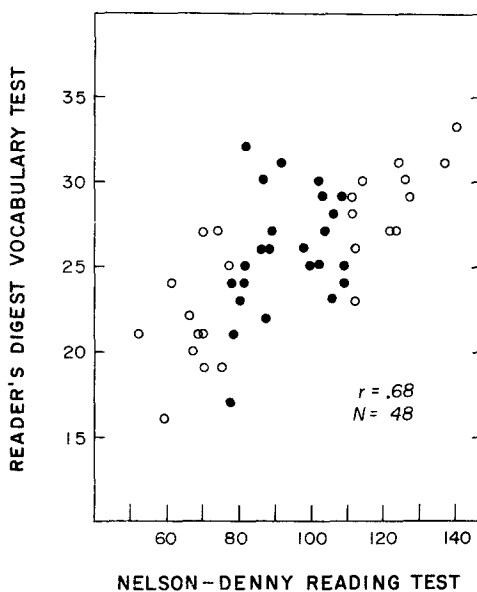


Fig. 4. Relationship between performance on the vocabulary test used by Butler and Hains (1979) and scores on the Nelson-Denny Reading Test, Form B, for each subject tested. Open circles indicate subjects used in Experiment 2.

Table 2. Intercorrelation matrix for 48 subjects on the vocabulary test and each part of the Nelson-Denny Reading Test

| Test                            | 1.  | 2.   | 3.   | 4.   |
|---------------------------------|-----|------|------|------|
| 1. Nelson-Denny (composite)     | 1.0 | 0.92 | 0.93 | 0.68 |
| 2. Nelson-Denny (vocabulary)    |     | 1.0  | 0.71 | 0.67 |
| 3. Nelson-Denny (comprehension) |     |      | 1.0  | 0.60 |
| 4. Vocabulary test              |     |      |      | 1.0  |

subjects clearly differed in terms of the vocabulary test but were selected with the Nelson-Denny Reading Test as the criterion.

### *Method*

*Subjects.* The initial participants were 48 volunteers from a second-year Psychology course at Queen's University. These students were administered the vocabulary test used by Butler and Hains (1979) and Form B of the Nelson-Denny Reading Test. Two groups of subjects were then formed by selecting 12 with the highest scores and 12 with the lowest scores on the reading test. These two groups were designated as Good Readers and Poor Readers, respectively. The terms "good" and "poor" are meaningful in a relative sense only and not in reference to population norms.

*Materials and Apparatus.* The stimuli consisted of 144 eight-letter, three-syllable words, 76 of which were among those used by Mewhort and Beal (1977). These words were divided into four sets of 36 each according to the serial position of the syllable boundaries. All syllable boundaries were determined using *Webster's New World Dictionary*, second college edition (1974). The four sets included: (1) syllable boundaries after 2 and 4 (e.g., va/ca/tion), (2) after 3 and 5 (e.g., hos/pi/tal), (3) after 2 and 5 (e.g., be/hav/ior), and (4) after 3 and 6 (e.g., con/tin/ue). These four sets of words were balanced for word frequency using the word frequency norms prepared by Kučera and Francis (1967).

In the syllable condition, each word was subdivided into three units corresponding to the syllables. In the nonsyllable conditions, the divisions for set (1) occurred after letters 3 and 5 (e.g., vac/at/ion), for set (2) after letters 2 and 4 (e.g., ho/sp/ital), for set (3) after 3 and 6 (e.g., beh/avi/or), and for set (4) after 2 and 5 (e.g., co/nti/nue).

A practice set of 20 eight-letter words was also prepared and similarly was divided for syllable and nonsyllable conditions. None of the practice words had syllable boundaries in the same positions as the words on the test list.

All stimuli were presented on a CRT display monitor (Tektronix model 604a) using a rapid decay P-4 phosphor. The monitor was controlled using a PDP 11/23 minicomputer with the n-channel tachistoscope algorithm designed by Mewhort (1978). All characters appeared as lower case letters by brightening the appropriate dots in a 25 × 13 dot matrix. The visual angle, subtended by one character, was approximately 22' by 22' while the row of eight letters subtended about 3°30' horizontally.

*Procedure.* Each subject received 144 trials, half with words divided into syllables and half divided into nonsyllables. One word was shown on each trial with the three units shown one at a time in succession. The order of presentation of the units was either left-to-right or right-to-left, while maintaining the correct spatial location for each unit. Each word unit was shown for 3 ms and the pause between each unit, that is the inter-unit interval or UII, was either 0, 250, or 500 ms. Thus, there were a total of 12 different presentation conditions for each subject and the subject viewed an equal number of stimuli under each condition. Across subjects, the words used in

each condition were counterbalanced so that each word appeared equally often in each condition.

Subjects were asked to fixate on a central dot prior to each trial and to initiate each trial using a remote control switch. Subjects were told to report the word verbally, to guess if they were uncertain, or to report any letters they had seen if they were unable to report a word. Subjects were given 20 practice trials prior to beginning the experiment.

### *Results and Discussion*

The results for each subject were first scored by counting the number of words that the subject failed to identify in each condition. These *word error* scores for all 24 subjects were then submitted to a  $2 \times 2 \times 3 \times 2$  analysis of variance to assess the effects of reading ability, type of unit, inter-unit interval, and direction of presentation. The word error score made no allowance for a partially correct response, for example, "location" for "vacation", nor did it allow an assessment of serial position or the temporal order of presentation. Therefore, two further scores were calculated. For the 0 IUI condition in which all three units seemed to appear simultaneously the probability of correctly reporting the letters for each stimulus position was calculated. Reporting the whole word correctly incremented the letter score for all eight positions but this permitted an assessment of partially correct word responses and letter report as well. For the 250 and 500 ms IUI conditions in which the pause between

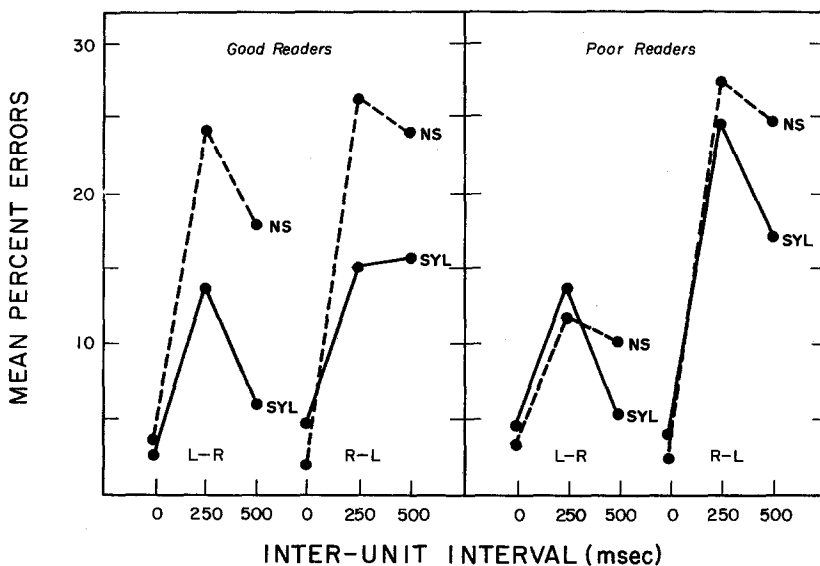


Fig. 5. Mean error rate for identifying words for each condition in Experiment 2.

units was quite apparent, the report of each single unit was assessed separately so that it was possible to examine the effects of temporal order.

*Word Errors.* The mean errors for word report are shown in Figure 5 and the basic results, for both Good and Poor readers, duplicate the results obtained by Mewhort and Beal (1977). Subjects made few errors (3.6%) in the 0 IUI condition, more in the 500 ms IUI condition (15.4%), and the greatest number of errors (19.8%), in the 250 ms condition,  $F(2,44) = 41.49$ ,  $p < 0.01$ . Subjects made more errors with right-to-left presentation (16%) than left-to-right (9.8%),  $F(1,22) = 24.34$ ,  $p < 0.01$ , but direction was relevant only for the two longer IUI conditions,  $F(2,44) = 6.02$ ,  $p < 0.01$ . For the 0 IUI condition, performance was the same whether units were shown left-to-right (3.6%) or right-to-left (3.5%), which confirms the phenomenological impressions that the separation of units was not noticeable at presentation rate.

The results show the same difference between the syllable and nonsyllable conditions found by Mewhort and Beal (1977). The error rate was 10.8% for syllable and 15% for nonsyllables,  $F(1,22) = 15.92$ ,  $p < 0.01$ , but the difference changed across the IUI conditions,  $F(2,44) = 7.37$ ,  $p < 0.01$ . The difference between syllables and nonsyllables was negligible at 0 IUI, 4.2% and 2.9% respectively, but was greater at 250 ms IUI (17% and 22.6%), and at 500 ms IUI (11.3% and 19.4%). The effect of unit type was essentially the same at the two longer IUI conditions,  $F(1,22) < 1.0$ . Again, this confirms that the units appeared to be simultaneous in the 0 IUI condition. Not only do the effects duplicate those obtained by Mewhort and Beal, but the actual means are very close to their results despite a change in the word sample and a shift from upper to lower case.

The most important aspect of the results shown in Figure 5 is the difference between Good readers and Poor readers. Overall the two groups did not differ in the accuracy with which they recognized the words,  $F(1,22) < 1.0$ . Good readers showed a mean error rate of 13.1% compared to only 12.7% for Poor readers. Good readers, however, were much more sensitive to the type of unit presented than were the Poor readers,  $F(1,22) = 5.42$ ,  $p < 0.05$ . Good readers averaged 9.8% errors with words in syllables and 16.4% with nonsyllables. In contrast, the Poor readers averaged 11.8% for syllables and 13.5% for nonsyllables. The Poor readers do not appear to be as sensitive to the type of unit presented but are much more sensitive than the Good readers to the direction of presentation,  $F(1,22) = 4.19$ ,  $p < 0.05$ . Poor readers made fewer errors with left-to-right presentations (9.5%) than with right-to-left presentations (17%), while the Good readers showed little difference between the same conditions (11.3% and 14.9%). Good readers, then, were strongly affected by the type of unit and less so by direction of presentation while Poor readers showed the opposite pattern.

These results can be explained quite simply by assuming that both groups must parse the letter sequences in order to recognize the words but that the size of the higher-order unit achieved during parsing differs for the two ability groups. Good readers appear to use the syllable as a unit. Poor readers, on the other hand, appear to deal with a unit smaller than the syllable so that they are not sensitive to unit type and far better with the nonsyllable condition. Since all the words were eight-letter words, the use of smaller units during parsing would mean that the Poor readers had

more units to deal with while recognizing each word and this would be especially critical for the reorganization that must occur with right-to-left presentation. Thus, the difficulty that Poor readers experienced with right-to-left presentation does not indicate that this group has a greater "left-to-right mindedness", in Anderson and Dearborn's (1952, p. 224) terms, but rather, that they have more items to reorganize and remember. This does predict that they should be more sensitive to the temporal order of presentation of the units. Therefore, the differential effects of unit type and direction of presentation can both be explained in terms of differences in the size of the unit used during letter parsing.

*Stimulus Position.* In the 0 ms IUI condition, all of the letters in the word were presented within 9 ms. Consequently, all the units seem to appear simultaneously. The interactions of unit type and direction with IUI bear this out. The 0 ms IUI condition, therefore, was the one condition in which it was reasonable to assess the accuracy for each of the eight individual letter positions in the word. A separate  $8 \times 2$  analysis of variance was conducted on the 0 ms IUI data above to test the effects of stimulus position and reading skill.

The results showed a strong effect of position,  $F(7,154) = 10.94$ ,  $p < 0.01$ , but no difference between reading groups,  $F(1,22) < 1.0$ , and no interaction between position and reading skill,  $F(7,154) < 1.0$ . Both Good and Poor readers showed the same error pattern; the error rate was 1% or less for the first six letters, that is, the left-most positions, then increased to 1.5% at position seven and 2.7% at position eight.

*Temporal Order of Units.* A more detailed analysis was also conducted on the data for the 250 and 500 ms IUI conditions. The scores in this analysis were the number of times each subject correctly reported each unit in each condition, either by pronouncing it as part of a word or by spelling it. The data were collapsed across the 250 and 500 ms IUI conditions and submitted to a  $3 \times 2 \times 2 \times 2$  analysis of variance to assess the effects of temporal order of presentation, direction, unit type, and reader group.

As the results in Figure 6 indicate, the temporal order of the word units had a substantial effect on performance,  $F(2,44) = 9.40$ ,  $p < 0.01$ . The mean error rate was highest for the first unit presented (12.5%), lower for the second unit (11.4%), and lowest for the last unit shown (9.3%). The effect of temporal order, however, depended on the direction of presentation,  $F(2,44) = 16.01$ ,  $p < 0.01$ . For the left-to-right presentations, order had virtually no effect, yielding means of 8%, 8.8%, and 8.9% but for right-to-left presentation temporal order yielded means of 18%, 13.9%, and 9.9% for the first, second, and third units, respectively. Subjects had considerable difficulty remembering the first unit shown, that is, the right-most unit in the word, when the direction of presentation was right-to-left.

The temporal order of the units was much more crucial for Poor readers than for Good readers,  $F(2,44) = 8.32$ ,  $p < 0.05$ . As Figure 6 indicates, Poor readers showed an effect of temporal order, regardless of whether the units were syllables or non-syllables. Good readers, on the other hand, were more sensitive to the type of unit presented,  $F(1,22) = 5.83$ ,  $p < 0.05$ , but showed a consistent error rate of about 12% for the first, second, and third units shown. Again, the results are quite

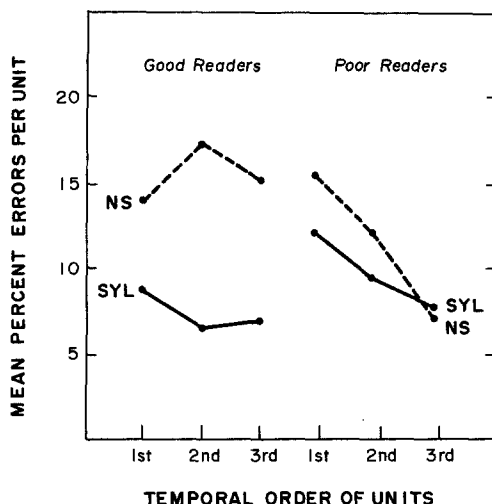


Fig. 6. Mean error rate for identifying each subword unit as a function of type of unit and temporal order in Experiment 2.

consistent with the interpretation that poor readers are using a smaller size unit in parsing and consequently have a greater load on short-term-memory.

## Conclusions

The results of these studies show substantial differences among adults in the use of orthographic information but, more importantly, the results also show how these differences can affect reading. In Experiment 1, subjects were asked to report briefly presented letter sequences varying in approximation to English. Subject scoring above the median on a vocabulary test were better able to utilize the redundancy in higher order approximations, compared to subjects scoring low on the test, but were hampered by the absence of redundancy with random letter sequences. This link between knowledge of vocabulary and the use of orthographic structure is similar to Mason's (1978) observation that skilled mature readers were more sensitive to orthographic structure than were less skilled readers. The results are contrary to Massaro and Taylor's (1980) results, however.

Experiment 2 extended the results of Experiment 1 in two ways: first, by showing that performance on the vocabulary test is closely correlated with reading and, second, by showing that adults differ substantially in their sensitivity to syllables as intraword units. This experiment used actual words, not pseudowords, presented as three temporally separate but spatially contiguous units. The word units either matched or violated syllable boundaries because previous studies by Mewhort and Beal (1977) and Taft (1979) have shown the importance of the syllable as a parsing unit. The results of the experiment showed that skilled readers are quite sensitive to the type of unit shown and made more errors with words divided into nonsyllable

units compared to syllable units. The less skilled readers, on the other hand, were more sensitive to the spatial direction and temporal order of the units than they were to the type of unit presented. If it is assumed that all subjects must parse a letter string prior to transferring the information to short-term memory (Mewhort & Campbell, 1981), then it would appear that skilled readers are more efficient than less skilled readers in using orthographic knowledge to parse the letters into larger higher-order units.

Coupled with the previous findings of Butler and Hains (1979), the present results provide an interesting profile of mature readers. Skilled readers have a good knowledge of vocabulary, derived probably from their reading experience, have a good understanding of orthographic structure, and have a processing strategy that focuses on the syllable during word recognition. Consequently, word recognition for these subjects appears to be "more wholistic" in that recognition performance is less affected by the number of letters in a word. Less skilled readers have less knowledge, both about vocabulary and about orthography, are less able to parse in syllable units, and must focus on an intraword unit smaller than the syllable. Hence the less skilled reader is more sensitive to word length in letters since this is a more accurate reflection of the number of units to be transferred to short-term memory. Given that word length is one determiner of fixation duration (Just & Carpenter, 1980; Thibadeau, Just, & Carpenter, 1982), these differences between good and poor readers should affect eye movement patterns in a dynamic reading task.

These differences between skilled and less skilled readers appear with younger children as well as adults. Scheerer-Neumann (1981) has reported a series of experiments with grade-three children that show that reading skill at that level is closely tied to the utilization of orthographic structure and sensitivity to the syllable as an intraword unit. Scheerer-Neumann found that an intervention program that taught children to segment words into syllables produced a substantial improvement for poor readers.

The fact that the same pattern of results emerges for both children and adults strongly suggests that an individual's reading skill is much more a result of a strategy adopted relatively early in learning to read rather than the extent to which one has progressed along a continuum of development. Good readers appear to be sensitive to orthographic structure early in their reading career and remain sensitive to it; poor readers may increase their reading performance by learning to process information more rapidly rather than more efficiently, much as an untrained typist can learn to "hunt and peck" quickly rather than "touch type" efficiently. Two readers at different grade levels could produce the same score on a reading test but the younger child may be using an efficient but slow strategy while the older child in the higher grade is using a relatively inefficient strategy that fails to utilize redundancy but is doing so faster.

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